Design Brief

AstroIllustrator | Consolidated specification v2.1 | 22 October 2026

1. Creative direction

Premium, scientific and editorial. Use restrained contrast, precise typography, consistent line weights and intentional whitespace. Distinguish true astronomical charts from conceptual teaching schematics and scenic composites.

2. Design tokens and presets

Dark Atlas: deep navy, magnitude-dependent star sizes, subdued cyan-gray constellation lines and white labels. Clean Scientific: neutral background, coordinate grid and minimal ornament. Educational Diagram: high-contrast geometric lines, angle arcs, clear legends. Print Monochrome: grayscale and accessible patterns. Scenic Composite: realistic star-free background plus locked accurate chart. A preset changes appearance, not scientific positions, measured angles or projection. Prefer vector SVG for atlas masters.

3. Scientific diagram grammar

Use consistent visual distinctions: solid line for horizon/axes, dashed line for reference meridian or zenith, curved arc for angles, dot/star glyph for target, thin leader lines for labels. Show degrees with degree symbol and explain altitude vs azimuth. Polaris diagram: observer at vertex, north horizon as baseline, line of sight to Polaris, highlighted altitude angle, vertical zenith reference and a note about approximate NCP vs exact Polaris. Do not confuse zenith angle with altitude angle: zenith distance = 90 degrees minus altitude.

4. Localization and captions

Provide sq and en terminology dictionaries: Zenith / Zeniti; Horizon / Horizonti; Altitude / Lartësia këndore; Azimuth / Azimuti; North Celestial Pole / Poli qiellor verior. Confirm Albanian editorial terminology before publication. Caption should include scene type, location/time where relevant, projection, approximations, data provenance and figure ID. Avoid crowding the image; use external captions when necessary.

5. Print and accessibility

Set target page size, bleed and safe margins via export profile. Keep label text as vectors where possible, embed/outline fonts only as licensing permits, test grayscale and 100% print size. Provide contrast-safe label hierarchy and alternate textual descriptions. For asterisms, use a different line style from constellation segments so the reader does not mistake asterism boundaries for IAU boundaries.

6. Visual QA checklist

Every angle arc agrees with numeric label; cardinal directions are correct; southern horizon is not mirrored; star positions do not change between styles; star labels do not overlap; decorative assets contain no fabricated stars; publication metadata is visible or embedded. Document control: This version supersedes earlier 1.0 and 1.1 scope descriptions. Keep requests, manifests, diagrams and AI integrations aligned with this specification.

v2.1 additions: global observer support, time-aware conjunctions and AstroIllustrator naming

Event illustration visual grammar

Include a true-west compass marker, clear horizon line, altitude scale or unobtrusive grid, labels for planets/Moon, and an optional measured separation arc. Distinguish calculated positions from artistic atmosphere; no decorative AI-generated planets or stars overlapping verified geometry.

Twilight visual presets

Add Western Twilight, Eastern Dawn, Planetary Pairing, Lunar Pairing and Event Comparison presets. Sunset glow must not suggest physically exact atmospheric brightness unless calibrated. Keep night/day contrast readable for print; show

date, location, local time, UTC offset, direction and field of view.

Scientific integrity in composition

Planet symbols may be intentionally enlarged for legibility, but must be marked not to scale. Lunar phase and orientation follow ephemeris-derived geometry. Preserve vector object layers, publication margins, text-safe areas and language-specific typography (Albanian and English).

Scientific note: illustrative dates and prompts are examples, not verified event predictions. All published charts require computed ephemerides and editorial review.